

LESSON 5-3 • HOMEWORK PACKET • SHEET 1 OF 2**Trapezoids by Decomposing**

Maryland Standard 6.G.A.1 • Learning target – I can find the area of a trapezoid by decomposing it into a rectangle and triangles.

REMEMBER

- **Decompose** a trapezoid into a rectangle and one or two triangles.
- Find the area of each piece, then **add** the areas.
- When you cut off a rectangle, the triangle bases together equal the **difference** of the two parallel sides.

WORKED EXAMPLE

Parallel sides 6 m and 12 m, height 5 m. Find the area.

Step 1 Rectangle: $6 \times 5 = 30$

Step 2 Triangle bases: $(12 - 6) \div 2 = 3$ m each

Step 3 Triangles: $2 \times (\frac{1}{2} \times 3 \times 5) = 15$

Step 4 Add: $30 + 15 = 45$

Answer **45 m²**

PRACTICE

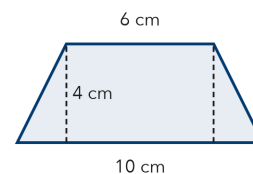
show your work • label units

1. The trapezoid is cut into a rectangle and two matching triangles.

a. Area of the rectangle _____ cm²

b. Area of one triangle _____ cm²

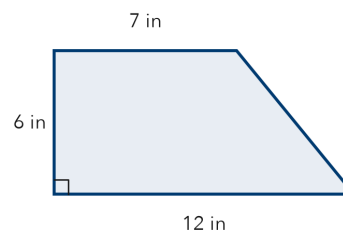
Area of trapezoid _____ cm²



2. Decompose the trapezoid into a rectangle and one triangle. Find its area.

Show your work.

Answer _____ in²



3. **Skate Ramp.** The side of a skate ramp is a trapezoid with one square corner. Its parallel sides are 3 ft and 8 ft, and its height is 4 ft. Find the area by decomposing.

Show your work.

Answer _____ ft²

4. **Table Top.** A table top is a trapezoid with parallel sides of 5 ft and 11 ft and a height of 3 ft. It is cut into a rectangle and two matching triangles. Find its area.

Answer _____ ft²

5. **Find the Error.** A trapezoid has parallel sides 9 cm and 15 cm and height 4 cm. Tomás says each of its 2 matching triangles has a base of 6 cm. Explain his mistake and find the area.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Find a trapezoid shape at home or outside (a lampshade side, a bag, a roof). Show someone how to cut it into a rectangle and triangles.

LESSON 5-3 • HOMEWORK PACKET • SHEET 2 OF 2

The Trapezoid Area Formula

Maryland Standard 6.G.A.1 • Learning target – I can find the area of a trapezoid using the trapezoid area formula.

REMEMBER

- $A = \frac{1}{2} \times (b_1 + b_2) \times h$. The bases b_1 and b_2 are the **parallel** sides.
- Add the bases first, then multiply by the height h , then take half.

WORKED EXAMPLE

Parallel sides 7 cm and 11 cm, height 6 cm. Find the area.

- Step 1** Add the bases: $7 + 11 = 18$
Step 2 Multiply by the height: $18 \times 6 = 108$
Step 3 Take half: $108 \div 2 = 54$
Answer 54 cm^2

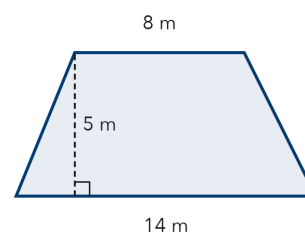
PRACTICE

show your work • label units

1. Find the area of the trapezoid.

Show your work.

Answer _____ m^2



2. Find the area of each trapezoid with bases b_1 and b_2 and height h .

- a. 4 ft, 10 ft; $h = 6 \text{ ft}$ _____ ft^2 b. 5 in, 9 in; $h = 8 \text{ in}$ _____ in^2
c. 6 m, 12 m; $h = 3 \text{ m}$ _____ m^2 d. 3 cm, 7 cm; $h = 9 \text{ cm}$ _____ cm^2

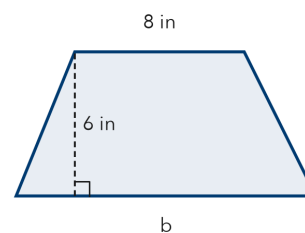
3. **Garden Plot.** A garden plot is a trapezoid with parallel sides of 16 ft and 24 ft and a height of 10 ft. One bag of mulch covers 25 ft^2 . How many bags are needed?

Answer _____ bags

4. **Missing Base.** The area of the trapezoid is 60 in^2 . Find the missing base b .

Show your work.

Answer _____ in



TEST PRACTICE

choose the best answer

5. A trapezoid has parallel sides of 6 cm and 10 cm and a height of 5 cm. What is its area?
A 21 cm^2 **B** 40 cm^2 **C** 80 cm^2 **D** 300 cm^2
6. Which expression finds the area of a trapezoid with bases 3 m and 9 m and height 4 m?
A $\frac{1}{2} \times 3 \times 9 \times 4$ **B** $(3 + 9) \times 4$ **C** $\frac{1}{2} \times (3 + 9) \times 4$ **D** $\frac{1}{2} \times 3 + 9 \times 4$

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Measure a trapezoid shape at home, like the side of a bag. Estimate its area with the formula.